

Chapter 99

Film Nodes

This chapter details the Film nodes in Fusion.

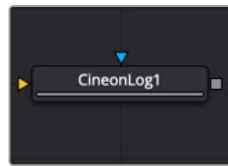
The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Cineon Log [Log]



The Cineon Log node

Cineon Log Node Introduction

The Cineon Log node is used to convert several different log camera formats to linear gamma and back again. Although the name implies that it should be used with Cineon files, it handles “log” gamma from many different digital cinema sources such as Blackmagic Design, Arri, and Red cameras.

Input

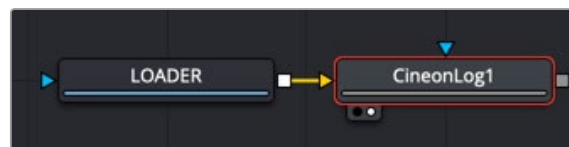
There are two Inputs on the Cineon Log node: one for the log image and one for the effects mask.

Input: The orange input is used for the primary 2D image that gets the highlight applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input restricts the log conversion to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

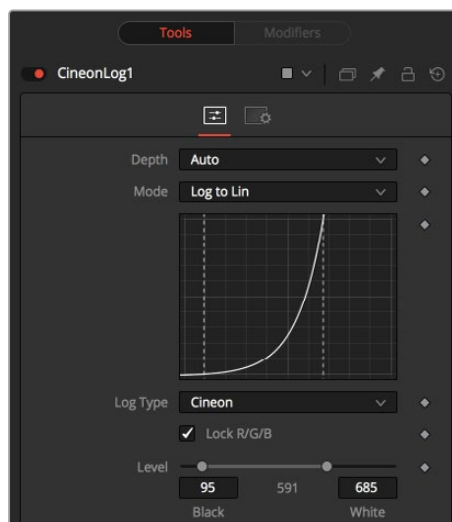
Basic Node Setup

The Cineon Log node is placed directly after a MediaIn node in DaVinci Resolve or a Loader node in Fusion Studio. It is also commonly placed before a MediaOut or Saver node to convert back to a Log-encoded image.



A Cineon Log node placed after a Loader node in Fusion Studio

Inspector



Cineon Log controls

Controls Tab

The Controls tab includes settings for converting from log gamma to linear or from linear to log. You first select the Mode and then the Log Type. For instance, choose Log to Lin from the Mode menu, and then select BMD Film if you are compositing with a RAW clip from a Blackmagic Design camera. Those settings output a linear image ready for compositing.

Depth

The Depth menu is used to select the color depth used to process the input image. The default option is Auto. Auto determines the color depth based on the file format loaded. For example, JPEG files automatically process at 8 bit because the JPEG file format does not store color depths greater than 8. Blackmagic RAW files load at Float, etc. If the color depth of the format is undetermined, the default depth defined in the Frame Format preferences is used.

Mode

The Mode menu offers two options: one for converting log images to linear and one for converting linear images to logarithmic.

Log Type

The Log Type menu allows you to select the source of the file. Typically, you select the camera used to create the image, although the Josh Pines option is specific to film scan workflows. This menu contains the following camera log types:

- Cineon
- Canon Log
- Panasonic V-Log
- Viper Film Stream
- Arri Log C
- Nikon N Log
- Red Log Film
- ACESlog
- BMD Film
- Panalog
- Sony S-Log

Lock RGB

When enabled, the settings in this tab affect all color channels equally.

Disable this control to convert the red, green, and blue channels of the image using separate settings for each channel.

Level

Use this range control to set the black level and white level in the log image before converting. The left handle adjusts the black level, while the right handle adjusts the white level. Pixels with values in log space below the black level become out-of-range values below 0.0. Pixels with values above the white level become out-of-range values above 1.0 after conversion.

When processing in floating-point color space, both negative and high out-of-range values are preserved. When using 16-bit or 8-bit mode, the out-of-range values are clipped.

Soft Clip (Knee)

The Soft Clip control is used to draw values that are out of range back into the image. This is done by smoothing the conversion curve at the top and bottom of the curve, allowing more values to be represented.

Applying a soft clip of any value other than 1 causes the node to process at 16-bit integer, eliminating all out-of-range values that do not fit within the soft clip.

Film Stock Gamma, Conversion Gamma, and Conversion Table

These controls are used to set the response curves of the logarithmic data during conversion. In addition to the settings above, a custom ASCII file Lookup Table (LUT) can be created with specific conversion values. The ASCII LUT file can be loaded using the Browse button.

Common Controls

Settings Tab

The Settings tab controls are common to all Film nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Black Rolloff

Since a mathematical $\log()$ operation on a value of zero or lower results in invalid values, Fusion clips values below $1e-38$ (0 followed by 38 zeros) to 0 to ensure correct results. This is almost never an issue, since values that small have no visual impact on an image. To see such tiny values, you would have to add three Brightness Contrast nodes, each with a gain set to 1,000,000. Even then, the values would hover very close to zero.

We have seen processes where instead of cropping these minimal values, they are instead scaled. So values between 0.0 and $1e-16$ are scaled between $1e-18$ and $1e-16$. The idea is to crush the majority of the visual range in a float image into values very near to zero, then expand them again, forcing a gentle ramp to produce a small ramp in the extreme black values. Should you find yourself facing a color pipeline using this process, here is how you can mimic it with the help of a Custom node.

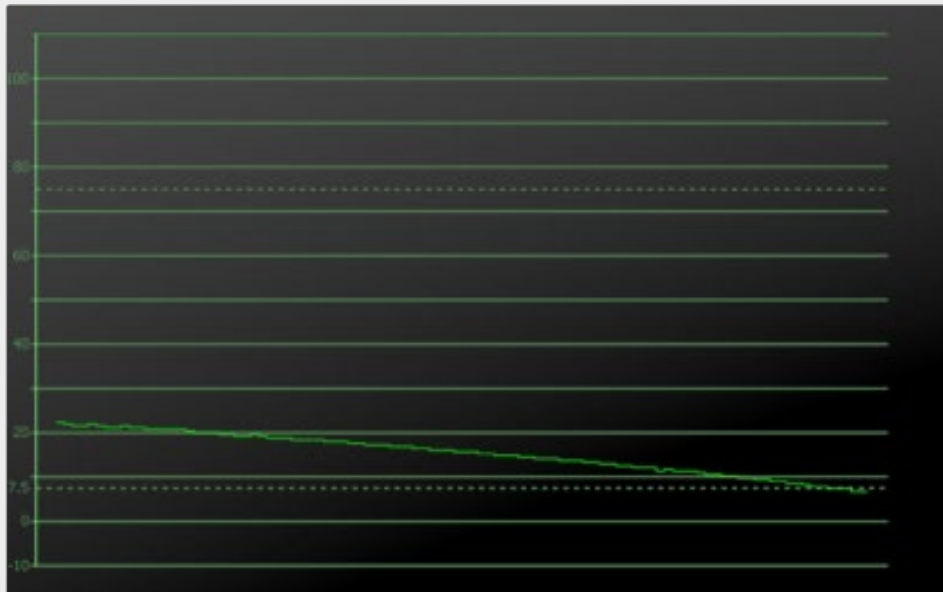
The process involves converting the log image to linear with a very small gamma and a wider than normal black level to white level (e.g., conversion gamma of 0.6, black of 10, white of 1010). This crushes most of the image's range into very small values. This is followed by a Custom node (described below), and then by a linear to log conversion that reverses the process but uses a slightly higher black level. The difference between the black levels defines the falloff range.

Since this lifts the blacks, the image is usually then converted back to linear one more time, using more traditional values (i.e., 95-685) to reset the black point.

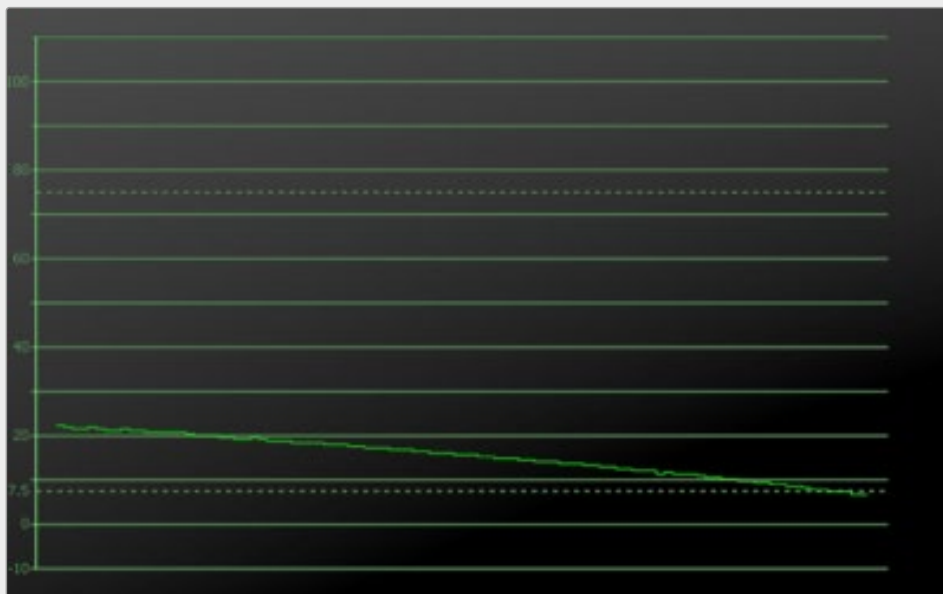
The Custom node should use the following equation in the red, green, and blue expressions:

```
if (c1< 1e-16, 1e-18 + (c1/1e-16)*(1e-16 - 1e-18), c1)
```

Falloff Comparison



The black falloff from the native Fusion process



Virtually identical black falloff from the ramped clipping process

Film Grain [FGr]



The Film Grain node

Film Grain Node Introduction

The Film Grain node adds generated film grain to an image. Typically, when compositing with different elements shot on film and in digital, the grain is removed for compositing operations and then reapplied to the final composite. This helps create the appearance that all elements were shot as a single scene with the same film stock.

NOTE: Although more accurate, the Film Grain node does not replace the older Grain node, which is still provided to allow older compositions to load and render, but in almost every case, it is better to use the Film Grain node.

Input

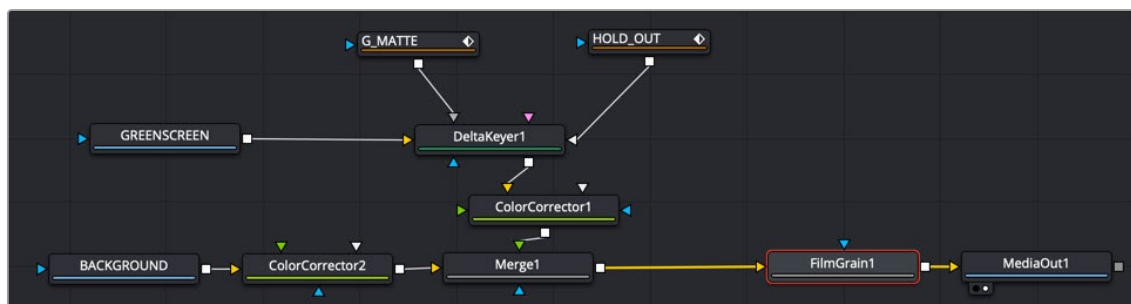
There are two inputs on the Film Grain node: one for the image and one for the effects mask.

Input: The orange input is used for the primary 2D image that gets the grain applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the grain to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

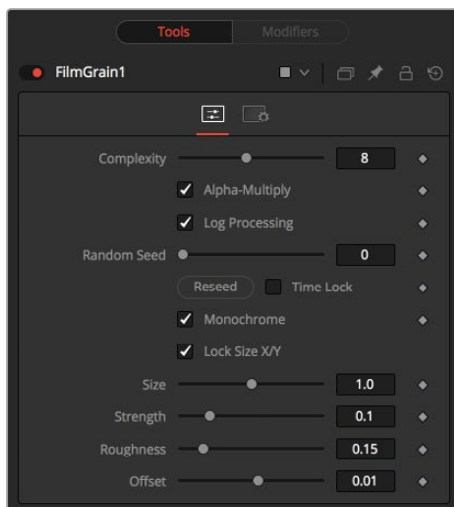
Basic Node Setup

The Film Grain node is commonly used before a MediaOut node in DaVinci Resolve or before a Saver node in Fusion Studio. Since grain is often removed from source material to achieve cleaner keys and composite, film grain is added back before the output to create a more realistic composite.



Film Grain node adding grain back before output

Inspector



Film Grain controls

Controls Tab

The Controls tab includes all the parameters for modifying the appearance of the film grain.

Complexity

The Complexity setting indicates the number of “layers” of grain applied to the image. With a complexity of 1, only one grain layer is calculated and applied to the image. When complexity is set to 4, the node calculates four separate grain layers and applies the mean combined result of each pass to the final image. Higher complexities produce visually more sophisticated results, without the apparent regularity often perceivable in digitally-produced grain.

Alpha Multiply

When the Alpha Multiply checkbox is enabled, the Film Grain node multiplies its results by the source image’s Alpha channel. This is necessary when working with post-multiplied images to ensure that the grain does not affect areas of the image where the Alpha is 0.0 (transparent).

NOTE: Since it is impossible to say what the final value of semitransparent pixels in the image are until after they are composited with their background, you should avoid applying log-processed grain to the elements until after they have been composited. This ensures that the strength of the grain is accurate.

Log Processing

When this checkbox is enabled (default), the grain applied to the image has its intensity applied nonlinearly to match the grain profile of most film. Roughly speaking, the intensity of the grain increases exponentially from black to white. When this checkbox is disabled, the grain is applied uniformly, regardless of the brightness of the affected pixel.

One of the primary features of grain in film is that the appearance of the grain varies radically with the exposure so that there appears to be minimal grain present in the blacks, with the amount and deviation of the grain increasing as the pixels exposure increases. In a film negative, the darkest portions of the developed image appear entirely opaque, and this obscures the grain. As the negative becomes progressively clearer, more of the grain becomes evident in the result. Chemical differences

in the R, G, B, layer's response to light also cause each color component of the film to present a different grain profile, typically with the blue channel presenting the most significant amount of grain.

As a result, an essential control in the Film Grain node is the Log Processing checkbox, which should be enabled when matching film, and disabled when working with images that require a more linear grain response. Having this checkbox enabled closely mimics the results of preceding the old Grain node with a Linear to Log conversion and following with a Log to Linear conversion immediately after.

Seed

The Seed slider and Reseed button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values produce the same random results. Click on the Reseed button to randomly select a new seed value, or adjust the slider to select a new seed value manually.

Time Lock

Enabling Time Lock stops the random seed from generating new grain on every frame.

Monochrome

When the Monochrome checkbox is enabled (default), the grain is applied to the red, green, and blue color channels of the image equally. When deselected, individual control over the Size, Strength, and Roughness of the grain in each channel becomes possible.

Lock Size X/Y

Deselect the Lock Size X/Y checkbox to control the size of the grain along the X- and Y-axis individually.

Size

The grain size is calculated relative to the size of a pixel. Consequently, changing the resolution of the image does not impact the relative appearance of the grain. The default grain size of 1.0 produces grain kernels that cover roughly 2 pixels.

Strength

Grain is expressed as a variation from the original color of a pixel. The stronger the grain's strength, the wider the possible variation from the original pixel value. For example, given a pixel with an original value of p , and a Grain node with complexity = 1 size = 1; roughness = 0; log processing = off; the grain produces an output value of $p \pm \text{strength}$. In other words, a pixel with a value of 0.5 with a grain strength of 0.02 could end up with a final value between 0.48 and 0.52.

Once again, that's a slight oversimplification, especially when the complexity exceeds 1. Enabling the Log Processing checkbox also causes that variation to be affected such that there is less variation in the blacks and more variation in the whites of the image.

NOTE: When visualizing the effect of the grain on the image, the more mathematically inclined may find it helps to picture a sine wave, where each lobe of the sine wave covers 1 pixel when the Grain Size is 1.0. The Grain Size controls the frequency of the sine wave, while the Grain Strength controls its amplitude. Again, this is something of an oversimplification.

Roughness

The Roughness slider applies low frequency variation to give the impression of clumping in the grain. Try setting the roughness to 0, and observe that the grain produced has a very even luminance variation across the whole image. Increase the roughness to 1.0 and observe the presence of "cellular" differences in the luminance variation.

Offset

The Offset control helps to match the intensity of the grain in the deep blacks by offsetting the values before the intensity (strength) of the grain is calculated. So an offset of 0.1 would cause a pixel with a value of 0.1 to receive grain as if its value was 0.2.

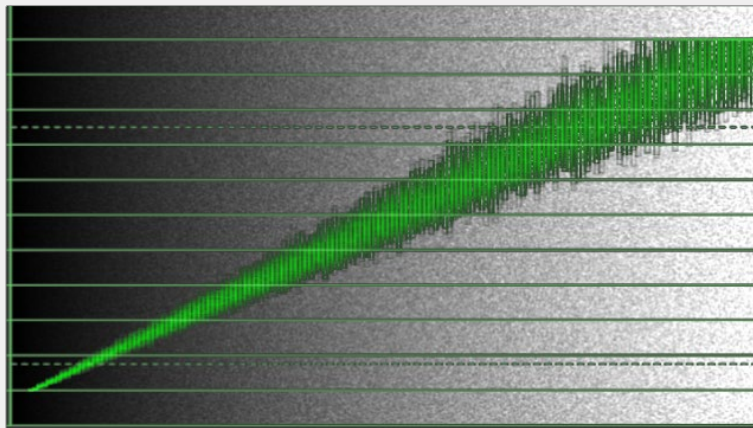
Common Controls

Settings Tab

The Settings tab controls are common to all Film nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

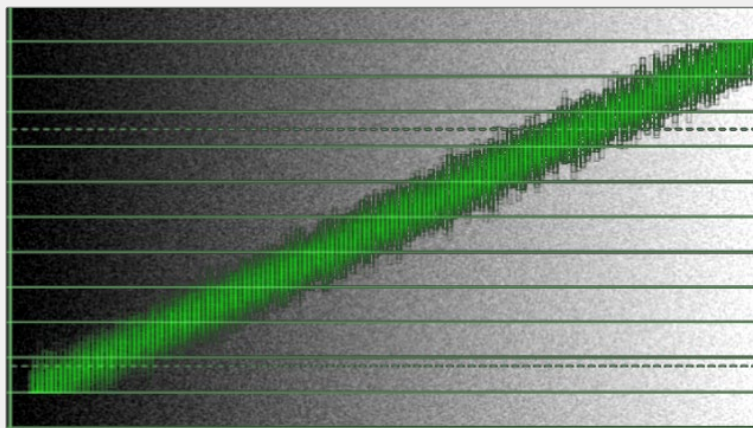
Processing Examples

Log Processing On



In the default setting, the different amounts of Grain are applied to the blacks and the whites of the image

Log Processing Off



When Log processing is off, the Grain is applied evenly to the entire image, as shown here

Grain [Grn]



The Grain node

Grain Node Introduction

The Grain node offers an older film grain emulation than the more modern and accurate Film Grain node. The Grain node is still provided to allow older compositions to load and render, but in almost every case, it is better to use the Film Grain node.

Input

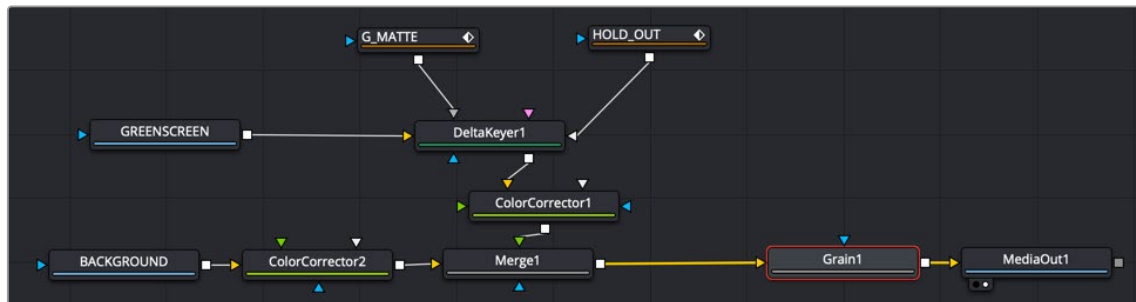
There are two Inputs on the Grain node: one for the 2D image and one for the effects mask.

Input: The orange input is used for the primary 2D image that gets the grain applied.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the grain to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

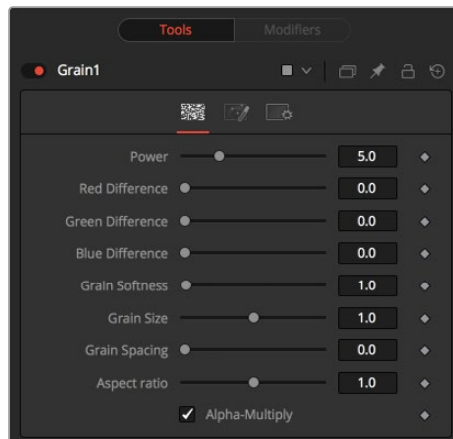
Basic Node Setup

The Grain node is commonly placed before a MediaOut node in DaVinci Resolve, or before a Saver node in Fusion Studio. It adds back grain previously removed in a composite.



A Grain node used to add grain back for a more realistic composite

Inspector



Grain controls

Controls Tab

The Controls tab includes all the parameters for modifying the appearance of the grain.

Power

This slider determines the strength of the grain. A higher value increases visibility, making the grain more prevalent.

RGB Difference

Separate Red, Green, and Blue sliders are used to modify the strength of the effect on a per channel basis.

Grain Softness

This slider controls the blurriness or fuzziness of the grain. Smaller values cause the grain to be more sharp or coarse.

Grain Size

This slider determines the size of the grain particles. Higher values increase the grain size.

Grain Spacing

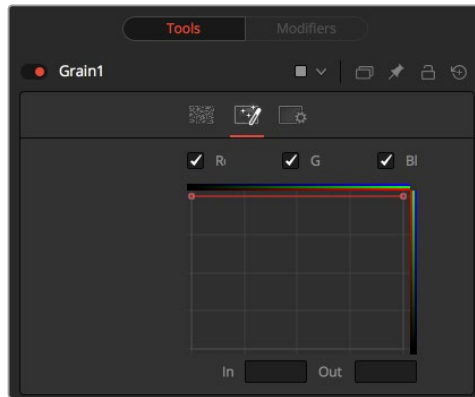
This slider determines the density or amount of grain per area. Higher values cause the grain to appear more spaced out.

Aspect Ratio

This slider adjusts the aspect of the grain so that it can be matched with anamorphic images.

Alpha-Multiply

When enabled, this checkbox multiplies the image by the Alpha, clearing the black areas of any grain effect.



Grain Spread controls

Spread Tab

The Spread tab uses curves for the red, green, and blue channels to control the amount of grain over each channel's tonal range.

RGB Checkboxes

The red, green, and blue checkboxes enable each channel's custom curve, allowing you to control how much grain appears in each channel. To mimic usual film responses, more grain would appear in the blue channel than the red, and the green channel would receive the least. Right-clicking in the spline area displays a contextual menu containing options related to modifying spline curves.

For more information on the LUT Editor's controls see *Chapter 107, "LUT Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 47 in the Fusion Reference Manual.

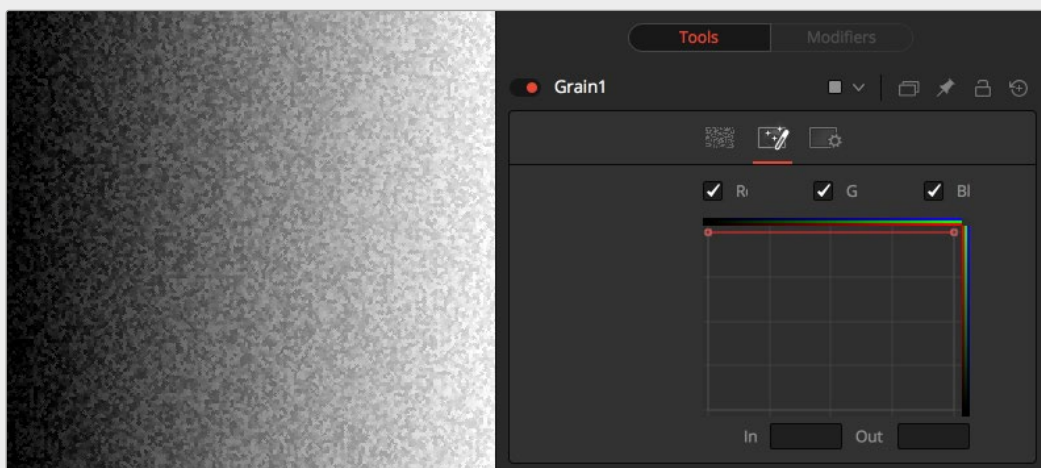
In and Out

This control provides direct editing of points on the curve by setting In/Out point values.

Examples

Default Spread

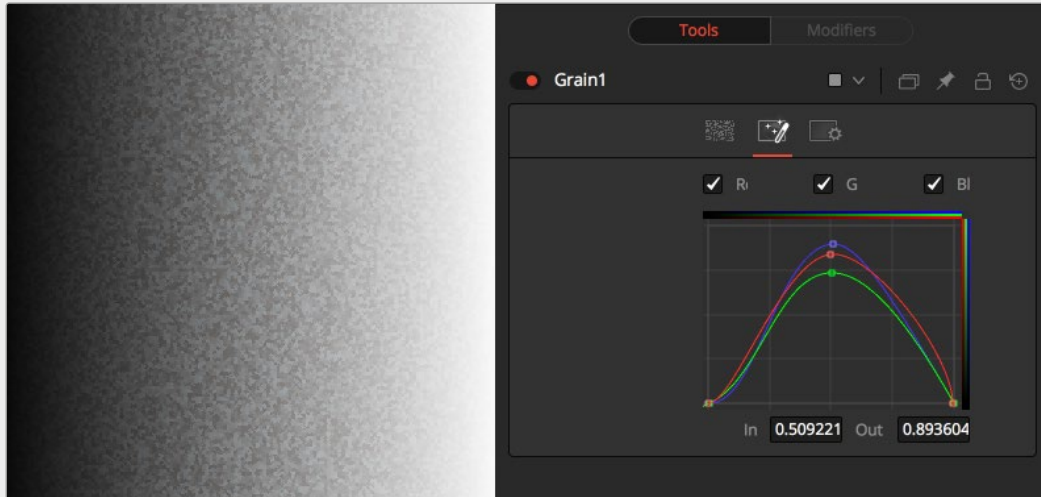
In the default setting, the grain is applied evenly to the entire image, as shown here. However, film often shows a different amount of grain in the blacks, mids, and whites.



Bell-Shaped Spread

Setting a bell shape is often a good starting point to create a more realistic-looking grain. Here we have a non-uniform distribution with different amounts of grain in the red, green, and blue channels.

In both examples, the grain's power has been exaggerated to show the effect a bit better.



Common Controls

Setting Tab

The Settings tab controls are common to all Film nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Light Trim [LT]



The Light Trim node

Light Trim Node Introduction

This node emulates film scanner light trims. By design, this node works best with logarithmic data, such as the images stored by Cineon, Arri, or Blackmagic RAW files. When logarithmic data is provided, the Light Trim node can be used to increase or decrease the apparent exposure level of the image.

Inputs

There are two Inputs on the Light Trim node: one for the 2D image and one for the effects mask.

Input: The orange input is used for the primary Log 2D image that gets its exposure adjusted.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the exposure change to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

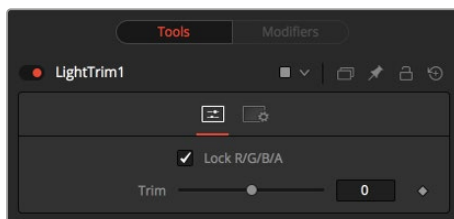
Basic Node Setup

The Light Trim node is placed after a LOG clip but before the Log clip is converted by a Cineon LOG node.



A Light Trim node used to adjust exposure on a LOG clip

Inspector



Light Trim controls

Controls Tab

The Controls tab includes a single slider that adjusts the exposure of the image.

Lock RGBA

When selected, the Lock RGBA control collapses control of all image channels into one slider. This selection is on by default. To manipulate the various color channels independently, deselect this checkbox.

Trim

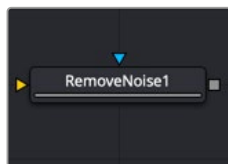
This slider shifts the color in film, optical printing, and lab printing points. 8 points equals one stop of exposure.

Common Controls

Settings Tab

The Settings tab controls are common to all Film nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

Remove Noise [RN]



The Remove Noise node

Remove Noise Node Introduction

The Remove Noise node provides simple noise management. The basic operation is that the node blurs the image channels, and then compares the blurred image to the original to extract the noise. A sharpness is then applied to the image, except where noise was detected.

To use this node, view the image and look at the red channel. Then increase the Red Softness until the grain appears to be gone. Next, increase the sharpness until the detail reappears, but stop before the grain reappears. Repeat for the green and blue channels.

Inputs

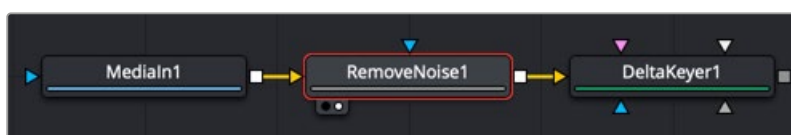
There are two inputs on the Remove Noise node: one for the 2D image and one for the effects mask.

Input: The orange input is used for the primary 2D image that gets noise removed.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the noise removal change to be within the pixels of the mask. An effects mask is applied to the tool after the tool is processed.

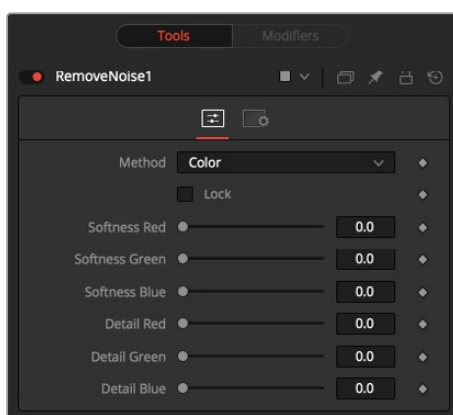
Basic Node Setup

The Remove Noise node can be used on any clip with noise. For example, it is used below to remove noise prior to keying the clip using the DeltaKeyer.



A Remove Noise node used to remove noise prior to keying

Inspector



Remove Noise controls

Controls Tab

The Controls tab switches the noise removal between two methods: Color and Chroma. When the Method is set to Color, the Controls tab adjusts the amount of blur and sharpness individually for each RGB channel. When the Method is set to Chroma, the blur and sharpness is adjusted based on Luma and Chroma controls.

Method

This menu is used to choose whether the node processes color using the Color or Chroma method. This also gives you a different set of control sliders.

Lock

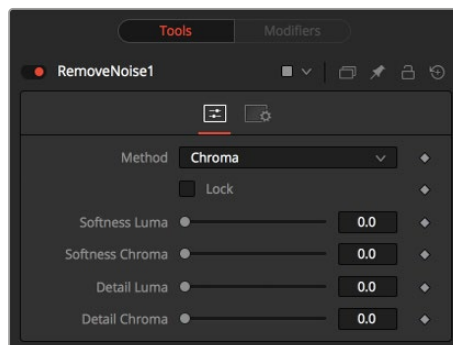
This checkbox links the Softness and Detail sliders of each channel together.

Softness Red, Green, and Blue

The Softness sliders determine the amount of blur applied to each channel of the image. In Chroma mode, you have sliders for the softness in the Luminance and Chrominance channels, respectively.

Detail Red, Green, and Blue

The Sharpness sliders determine how much detail is reintroduced into each channel after each channel is softened. In Chroma mode, you have sliders for Luminance and Chrominance channels, respectively.



Chroma Method controls

Common Controls

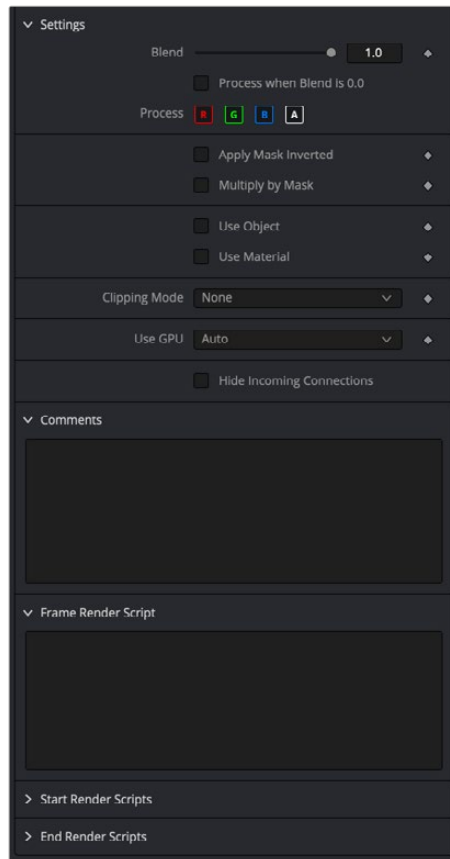
Settings Tab

The Settings tab controls are common to all Film nodes, so their descriptions can be found in the following “The Common Controls” section.

The Common Controls

Film nodes share a number of identical controls in the Inspector. This section describes controls that are common among Film nodes.

Inspector



The Common Film settings inspector

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Film category. The Settings controls are even found on third-party Film-type plugin tools. The controls are consistent and work the same way for each tool, although some tools do include one or two individual options, which are also covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Commonly, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Clipping Mode

This option determines how edges are handled when performing domain of definition rendering. This is mostly important for nodes like Blur, which may require samples from portions of the image outside the current domain.

- **Frame:** The default option is Frame, which automatically sets the node's domain of definition to use the full frame of the image, effectively ignoring the current domain of definition. If the upstream DoD is smaller than the frame, the remaining area in the frame is treated as black/transparent.
- **Domain:** Setting this option to Domain respects the upstream domain of definition when applying the node's effect. This can have adverse clipping effects in situations where the node employs a large filter.
- **None:** Setting this option to None does not perform any source image clipping at all. This means that any data required to process the node's effect that would normally be outside the upstream DoD is treated as black/transparent.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.